



MINING DEALER BEST PRACTICE

Upgrade Pulley Carrier MD6420-16

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

Upgrade Pulley Carrier MD6420-16	1
1.0 Introduction	2
2.0 Best Practice Description	2
3.0 Implementation Steps	3
4.0 Benefits	6
5.0 Resources Required	6
6.0 Supporting Attachments / References	6
7.0 Related Best Practices	6
8.0 Acknowledgements	7

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August 2023
1.01-230802-1

1.0 Introduction

The MD6420-16 drilling rigs in operation at the Conceição Mine use the PN467-7871 roller to guide the pulley carrier. These rollers have been constantly breaking during operation and falling from the mast reaching equipment components.

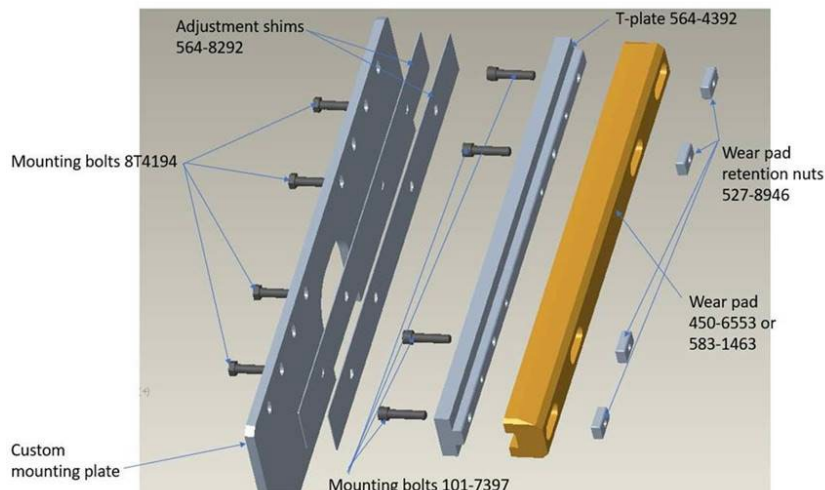
The Roller pulley carrier guides presented the first failure after approximately 6000 hours of operation and after this failure, consecutive failures have begun in this component. In some cases, the roller failure has resulted in damage to beam I which has the function of along with rollers keep the pulley carrier aligned.

After checking the fixing, roller alignment, adjustment of pulldown cables and correct adjustment of the auto tensioner of the pulldown cables, it was identified that the failures of the roller guides of the pulley carrier were still recurrent.

Last year we received two MD6310 in Brazil and we realized that the guide system of the pulley carrier was different from the MD6420-16 equipment. In view of this, The Sotreq engineering team carried out a study and realized that it would be possible to install the sliding guide system used in the MD6310 in the MD6420-16 equipment, making some adaptations.

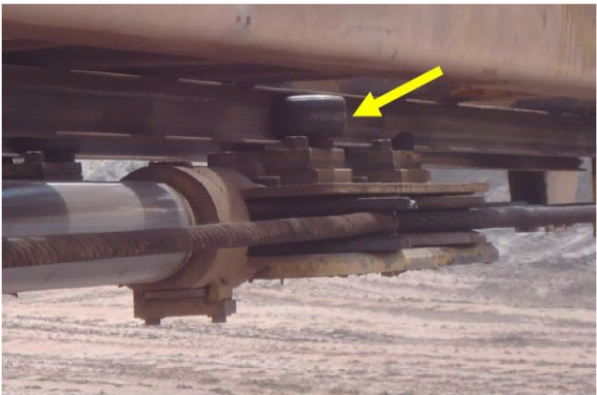
2.0 Best Practice Description

The Project consists of the replacement of the roller guides of the pulley carrier currently installed in the equipment by a set of sliding guides that uses polyurethane guides.

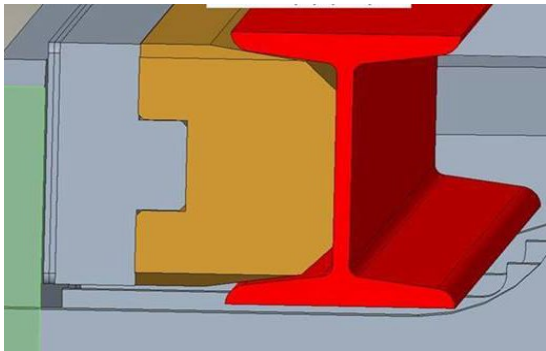
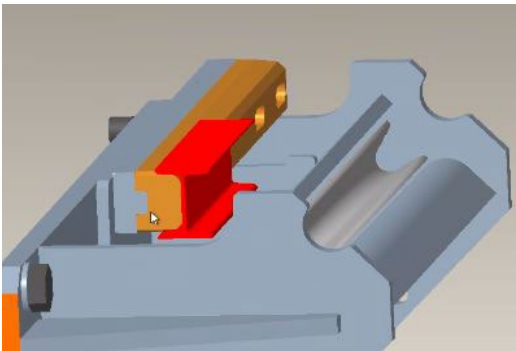


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Improvement installation location



Improvement to be implemented



3.0 Implementation Steps

Installation process

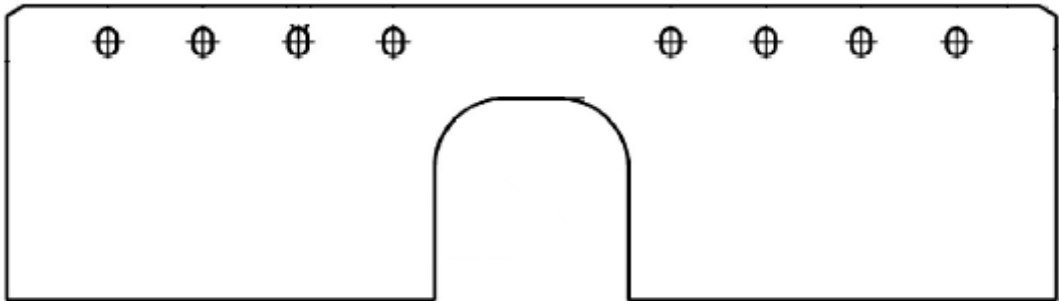
Inspection of beams I that received the sliding guides. If they are damaged replacement of the beams will be required.

To install the improvement, it is necessary to rework the support 527-8190 of the sliding guides as shown in the figure below.

After reworking the support, it needs to weld in the guide carrier and install the other parts. all other items is installed similar of the MD6310 equipment.

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Upgrade Guide Carrier Pulley MD6420	DATE 8/2/2023	CHG NO 00	NUMBER 1.01-230802-1

Original Support 527-8190



PN: 527-8190

Reworked Support - 527-8190



Technical Drawing on attachment: Reworked Support Carrier Technical Standard Drawing - MD6420-16 rev

Improvement in installation process



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Upgrade Guide Carrier Pulley MD6420	DATE 8/2/2023	CHG NO 00	NUMBER 1.01-230802-1
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Estimated time of application of the set of 24 hours of work, 3 days.

Note: To perform the task it is recommended to remove the mast for greater agility and ease of services to be performed.

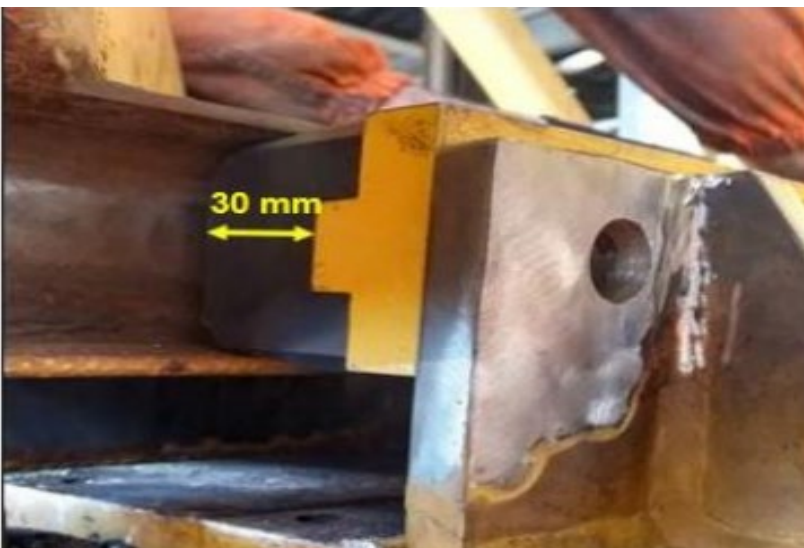
Care for release of equipment after installation of the project

Check the conditions of Hoist and pulldown cables, if they are near of end lifespan replace them. Perform inspections on Hoist and pulldown cables in the 250-hour equipment preventive maintenance.

Check pressure adjustment of the auto tensioning system of the pulldown cables and after release perform check every 500 hours of operation.

Inspection

For the project, it was defined that inspection would be necessary every 250 hours of operation. This inspection consists of checking the wear of the sliding shim and the fixing of all components present in the new guide installed.



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Upgrade Guide Carrier Pulley MD6420

DATE
8/2/2023

CHG
NO
00

NUMBER
1.01-230802-1

4.0 Benefits

The use of the new guides results:

In a greater contact with the I-beam, thus distributing the load and unloaded force on the beam.
 Increasement MTBR
 Better operation security
 Decrease the risk of parts falling

5.0 Resources Required

P/N	Description	LEAD TIME	Qnt for Machine	DEALER NET UNID COST	DEALER NET TOTAL COST
5648292	Adjustment shims	30 Days	8	R\$ 267,89	R\$ 2.143,12
8T4194	Bolts	Immediate	16	R\$ 10,36	R\$ 165,76
1017397	Bolts	3 Days	16	R\$ 46,69	R\$ 747,04
5644392	T-plate	30 Days	4	R\$ 10.166,15	R\$ 40.664,60
5278946	Retention nuts	30 Days	16	R\$ 283,68	R\$ 4.538,88
5831463	Wear pad	90 days	4	R\$ 1.432,66	R\$ 5.730,64
5278190	Plate	Manufactury locally	4	R\$ 1.870,00	R\$ 7.480,00
Total			68		~12.800 USD

Its necessary 1 mechanic technician and 1 welder.
 Estimated application time of the set 24 hours, 3 days.

Note: The plate 527-8190 need to be rework at field.

6.0 Supporting Attachments / References

- Reworked Support Carrier Technical Standard Drawing - MD6420-16
- Parts list

7.0 Related Best Practices

N/A

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Upgrade Guide Carrier Pulley MD6420	DATE 8/2/2023	CHG NO 00	NUMBER 1.01-230802-1

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